Young Woo Choi

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Education

Ph.D. in Physics Mar 2016 – Feb 2021
Department of Physics, Yonsei University, Seoul, Korea
(Advisor: Prof. Hyoung Joon Choi)

B.S. in Physics Mar 2010 – Feb 2016
Department of Physics, Yonsei University, Seoul, Korea
(Mandatory military service from Oct 2012 to Jul 2014)

Experience

Assistant Professor Sep 2024 – present
Department of Physics, Sogang University, Seoul, Korea

Postdoctoral Researcher Jun 2021 – Aug 2024
Department of Physics, University of California, Berkeley, CA, USA
(Advisor: Prof. Marvin L. Cohen)

Postdoctoral Researcher Mar 2021 – May 2021
Department of Physics, Yonsei University, Seoul, Korea
(Advisor: Prof. Hyoung Joon Choi)

Research Interests

- **Emergent phenomena in correlated and low-dimensional quantum materials**
  - Electron-phonon coupling, polarons, superconductivity, charge-density waves, and excitons
- **Design of novel quantum materials and nanostructures**
  - Moiré superlattices, electrides, 1D nanostructures, molecular junctions, and 2D heterostructures
- **Development of first-principles electronic-structure methods**
  - Density functional theory, many-body perturbation theory, and AI-assisted computational workflows

Research Highlights

- 24 journal articles published or accepted, in journals including Science, Nature Communications, Physical Review Letters, ACS Nano, and Advanced Materials.
- Google Scholar metrics: 514 citations; h-index 9. (29 Oct 2026)

Journal Publications

† Equal contributions; * Corresponding author.

[24] Min-Seok Kim†, Dong Cheol Lim†, Woo-Yong Kim, Sang Yong Song, Heejeong Koo, Moon-Ho Jo, **Young Woo Choi***, Yeongkwan Kim*, Jungpil Seo*, and Sung Wng Kim*,
Real-space imaging and manipulation of single electrons on a solid surface,
Science (accepted, 2026).

[23] Yoonseok Oh†, Changmo Kang†, Min Chul Choi†, Jaewoo Choi, Sung Won Jung, Timur K. Kim, Cephise Cacho, Makoto Hashimoto, Donghui Lu, Patrick S. Kirchmann, Jonathan A. Sobota, Zhi-Xun Shen, **Young Woo Choi***, and Keun Su Kim*,
Multivalley polarons at the onset of a Lifshitz transition in doped WS_2,
Nature Communications (accepted, 2026).

[22] YoungJu Park†, Taegil Kim†, Jonghyeok Baek†, Seongbin Park, Daeho Kim, ChaeHo Shin, Sam Sokhuoy, Wonyoung Kim, Sun Dongbang, Doseok Kim, **Young Woo Choi***, and Junwoo Park*,
Non-equilibrium Resonance Scattering in Intramolecular Charge Transport Drives Conformational Changes,
Nature Communications (accepted, 2026).

[21] Yuanyuan Qiu†, Bin Hua†, Zhenping Wang†, Fei Hu†, Dong Zhang, Yang Lu, Marvin L. Cohen, Alex Zettl, Kecheng Cao, Cong Su*, **Young Woo Choi***, and Qingqing Ji*,
Boltzmann Sigmoidal Modulation of Vanadium Doping in Monolayer MoS_2 by Potassium Iodide,
Nano Letters 26, 13836 (2026).

[20] Kimoon Higashihira Han†, Dongju Hwang†, Dongsin Kim†, Yeojin Ahn, Jaehun Cha, Seonggeon Gim, Gyubin Lee, Mingi Jho, Chan-young Lim, Sae Hee Ryu, Chris Jozwiak, Aaron Bostwick, Eli Rotenberg, Jonathan D. Denlinger, Makoto Hashimoto, Donghui Lu, **Young Woo Choi***, Gil Young Cho*, and Yeongkwan Kim*,
Unexpected stabilization of a single-q charge density wave in pristine 1T-$TiSe_2$,
Nature Communications (2026). doi:10.1038/s41467-026-78072-0.

[19] Tae Keun Yun†, Byeongchan Lee†, Soyeong Kwon, Jieun Yeon, Doohee Cho, **Young Woo Choi***, Kwanpyo Kim*, and SungWoo Nam*,
Anisotropic Γ-Valley Interlayer Excitonic Emission in Anisotropic-Isotropic 2D Heterostructures,
ACS Nano 20, 26215 (2026).

[18] Byoungkyu Kim, Younsung Chang, Changjun Lee, **Young Woo Choi**, and Soyeun Kim*,
Temperature-dependent spectroscopic ellipsometry study of exciton–phonon coupling in bulk $ReSe_2$,
Current Applied Physics 92, 37 (2026).

[17] Byunghyun Kim†, Ayoung Yuk†, Yunyeong Chang†, Dongsin Kim†, Daesung Park, **Young Woo Choi***, Miyoung Kim*, Moon Sung Kang*, and Hyobin Yoo*,
Dislocation Networks in Marginally Twisted Bilayer MoS_2,
ACS Materials Letters **8**, 1382 (2026).

[16] Byeongin Lee, Junho Bang, Sayan Banerjee, João Augusto Sobral, **Young Woo Choi**, Claudia Felser, Mathias S. Scheurer, Jian-Feng Ge, and Doohee Cho,
Anomalous impurity-induced charge modulations in black phosphorus,
Physical Review **B** 113, L041105 (2026).

[15] Yongjae Cho, Sushanta Pal, Stephen J. Buffat, Hao-Yu Lan, **Young Woo Choi**, Dmitry Y. Zemlyanov, Connor Devitt, Sunil A. Bhave, Zhihong Chen, and Joerg Appenzeller,
Extension p-Doping of Carbon Nanotube Transistors through Nitric Oxides Annealing,
ACS Nano **19**, 39014 (2025).

[14] **Young Woo Choi**†, Yangjin Lee†, Kwanpyo Kim, Alex Zettl, and Marvin L. Cohen,
Atomic and Electronic Structures of 1D Phosphorus Nanoring and Nanohelix,
ACS Nano **19**, 12155 (2025). († Equal contributions)

[13] **Young Woo Choi**, Jisoon Ihm, and Marvin L. Cohen,
Pairing interaction from three-dimensional acoustic plasmon demon modes in Sr_2RuO_4,
Physical Review B **110**, 155127 (2024).

[12] Yangjin Lee†, **Young Woo Choi**†, Linxuan Li, Wu Zhou, Marvin L. Cohen, Kwanpyo Kim, and Alex Zettl,
SiX_2 (X=S, Se) Single Chains and (Si-Ge)X_2 Quaternary Alloys,
ACS Nano **18**, 17882 (2024). († Equal contributions)

[11] Jeehong Park†, Soonsang Huh†, **Young Woo Choi**†, Donghee Kang, Minsoo Kim, Donghan Kim, Soohyung Park, Hyoung Joon Choi, Changyoung Kim, and Yeonjin Yi,
Visualizing the low-energy electronic structure of prototypical hybrid halide perovskite through clear band measurements,
ACS Nano **18**, 7570 (2024). († Equal contributions)

[10] Yangjin Lee†, **Young Woo Choi**†, Kihyun Lee, Chengyu Song, Peter Ercius, Marvin L. Cohen, Kwanpyo Kim, and Alex Zettl,
One-dimensional Magnetic MX_3 Single-Chains (M=Cr, V and X=Cl, Br, I),
Advanced Materials **35**, 2307942 (2023). († Equal contributions)

[9] Soo Yeon Lim†, Han-gyu Kim†, **Young Woo Choi**†, Takashi Taniguchi, Kenji Watanabe, Hyoung Joon Choi, and Hyeonsik Cheong,
Modulation of phonons and excitons due to moiré potentials in twisted-bilayer of $WSe_2/MoSe_2$,
ACS Nano **17**, 13938 (2023). († Equal contributions)

[8] Yangjin Lee†, **Young Woo Choi**†, Kihyun Lee, Chengyu Song, Peter Ercius, Marvin L. Cohen, Kwanpyo Kim, and Alex Zettl,
Tuning the Sharing Modes and Composition in a Tetrahedral GeX_2 (X = S, Se) System via One-Dimensional Confinement,
ACS Nano **17**, 8734 (2023). († Equal contributions)

[7] Jangwon Kim†, Youjin Lee†, **Young Woo Choi**†, Taek Sun Jung, Suhan Son, Jonghyeon Kim, Hyoung Joon Choi, Je-Geun Park, and Jae Hoon Kim,
Terahertz spectroscopy and DFT analysis of phonon dynamics of the layered van der Waals semiconductor Nb_3X_8 (X = Cl, I),
ACS Omega **8**, 14190 (2023). († Equal contributions)

[6] **Young Woo Choi** and Marvin L. Cohen,
Resonantly Enhanced Electromigration Forces for Adsorbates on Graphene,
Physical Review Letters **129**, 206801 (2022).

[5] **Young Woo Choi** and Hyoung Joon Choi,
Dichotomy of Electron-Phonon Coupling in Graphene Moiré Flat Bands,
Physical Review Letters **127**, 167001 (2021).

[4] **Young Woo Choi** and Hyoung Joon Choi,
Anisotropic Pseudospin Tunneling in Two-Dimensional Black Phosphorus Junctions,
2D Materials **8**, 035024 (2021).

[3] **Young Woo Choi** and Hyoung Joon Choi,
Intrinsic band gap and electrically tunable flat bands in twisted double bilayer graphene,
Physical Review B **100**, 201402(R) (2019).

[2] **Young Woo Choi** and Hyoung Joon Choi,
Role of Electric Fields on Enhanced Electron Correlation in Surface-Doped FeSe,
Physical Review Letters **122**, 046401 (2019).

[1] **Young Woo Choi** and Hyoung Joon Choi,
Strong electron-phonon coupling, electron-hole asymmetry, and nonadiabaticity in magic-angle twisted bilayer graphene,
Physical Review B **98**, 241412(R) (2018).

Preprints

[P3] Benran Zhang, Byeongchan Lee, Zien Zhu, Chih-En Hsu, Zhenfa Zheng, Hung-Chung Hsueh, **Young Woo Choi**, Mauro Del Ben, and Zhenglu Li*,
Enabling large language model agents for high-fidelity many-body perturbation theory calculations,
arXiv:2610.20779 (2026).

[P2] Sahiba Bano, Ghita Khatiri, Mitisha Jain, Prince Sharma, Saurabh Saini, Byeongchan Lee, **Young Woo Choi**, Zhenglu Li, Rajiv Kumar Singh, Soumitra Satapathi, Satoshi Ishii, Marios Zacharias, and Francisco Palazón,
Polymorphous Structural Landscapes Drive Broadband Optical Response in Water-Stable Mixed-Halide Antiperovskites,
ChemRxiv, doi:10.26434/chemrxiv.15008906/v1 (2026).

[P1] Kawon Lee, Sangeun Oh, **Young Woo Choi***, and Jeong-Hyuck Park*,
Attractive statistical forces and Pauli crystal formation in trapped Fermi gases,
arXiv:2605.12043 (2026).

Research Grants

- Nano and Materials Technology Development Program, National Research Foundation of Korea, 2026–present (Sogang University PI; programmable nanotube filling for angstrom-scale devices)
- Sogang G-LAMP Research Grant, Sogang University, 2024–present (overall program: KRW 5B/year, approx. USD 3.3M/year, for six years; Choi allocation: ~KRW 160M/year, approx. USD 106k/year)
- Post-doctoral Training Grants, National Research Foundation of Korea, 2023 (KRW 45M/year; approx. USD 30k/year)
- Global Ph.D. Fellowship, National Research Foundation of Korea, 2017–2021 (KRW 30M/year; approx. USD 20k/year)

Teaching Experience

Machine Learning Applications in Physics	Spring 2026, Spring 2025
Computational Physics II	Fall 2026
Computational Physics	Fall 2025
General Physics I	Spring 2026, Spring 2025
General Physics II	Fall 2026, Fall 2025
Statistical Physics	Fall 2024